

21CSC203P — ADVANCED PROGRAMMING PRACTICE

Unit – 1 : Session – 4 : SLO – 1 | Define Paradigms

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Activity (Solutions)

- 1) Procedural: program = procedures operating on data — `static int add(int a, int b){ return a+b; }` OOP: program = objects with state+behavior — `class Calculator { int add(int a,int b){ return a+b; } }` Functional: computation = evaluation of pure functions — `IntBinaryOperator add = (a, b) -> a + b;`
- 2) Same logic both ways (area of rectangle) — Procedural: `static double area(double l, double w){ return l*w; }` called as `area(4,5)`. Object-oriented: `class Rect { double l,w; Rect(double l,double w){this.l=l;this.w=w;} double area(){ return l*w; } }` called as `new Rect(4,5).area()`.
- 3) Java features supporting multiple paradigms: classes/inheritance (OOP), static methods and control structures (procedural/imperative), lambda expressions, method references, functional interfaces and the Stream API (functional), and annotations/generics (declarative flavor).